DRAFT — PROJECT QUANTUM COMP: GEMINI REQUEST PACKET, ALL TWELVE AGENTS

Draft request packet, NOT validated findings. Compiled November 5, 2026, finalized at about 15:25 CST (America/Chicago), by document finalization only.

What this is: the full, verbatim text of the twelve agents' fresh Gemini request intakes (questions, needed definitions, proof targets, open problems, and stated resolutions), each under its agent heading with its source file and SHA256. It also has a short governing preamble and clearly marked additive precision notes. Nothing in the agent sections has been rewritten; the additive notes are labeled "ADDITIVE PRECISION NOTE (compiler)" or "ADDITIVE CORRECTION" and sit directly above the verbatim text they qualify. A separate manifest pins all hashes.

A. GOVERNING PREAMBLE

A1. Research objective. The S²-11DM²ET-X general-relativity extension must include and recover the ENTIRE Standard Model. This is the Owner's objective; it has NOT been achieved or proven. Keep these two tiers apart:

(i) Minimal Standard Model: gauge group SU(3)×SU(2)×U(1)_Y, three generations of chiral fermions with hypercharges, anomaly cancellation, Higgs doublet and electroweak symmetry breaking to U(1)_em, Yukawa couplings and CKM mixing, couplings and RG running, and correspondence with tested SM observables.

(ii) Beyond-minimal checks: nonzero neutrino masses and PMNS mixing, dark matter, dark energy, baryogenesis, and other cosmological correspondence tests.

(Source: OWNER_DIRECTIVE_GR_EXTENSION_FULL_STANDARD_MODEL_RECOVERY_2026-11-05.md, SHA256 6237cc716829c2838368099382c7ecdebfed46895491f8ecb37ee5cc167c815e.)

A2. Immediate research halt. At about 15:17 CST on November 5, 2026, the Owner gave a direct instruction, relayed from the live call: an immediate, full computational-research halt, regardless of usage. The halt was confirmed to have reached all twelve agents. This is a summary of the Owner's instruction, not a transcript citation. The halt supersedes the primary autonomy rule (bcc3ced0f1dfe15560e0bdb2e45bb35f4b1db26dcfb439e647a98c12fc5bc9a9) and comes ahead of the 80-percent threshold. Until the Owner resumes work, there is no new scientific computation, research, or Gemini query. Only safe finalization of existing material is allowed. That includes THIS PDF, which the Owner expressly requested at about 15:20 CST.

A3. Weekly usage stop. Under the standing 80-percent rule, an observed weekly usage of 80 percent or more stops new research and computation. After that, only bounded finalization is allowed: collect full final reports from all twelve agents and consolidate them for the Owner. Work does not resume automatically after the weekly reset; it resumes only when the Owner authorizes it. Usage is checked at work checkpoints; it is not a continuously enforced automatic cap. (Source: OWNER_DIRECTIVE_WEEKLY_USAGE_80_STOP_AND_FINAL_REPORT_2026-11-05.md, SHA256 51ff16771c69d877675ac7c59e269d76444180784e81842c7818029387d1747e.) The last direct usage check was 60.9 percent weekly, shared across apps, at about 15:13 CST. That reading is not current.

A4. Gemini workflow, when authorized. Ask Gemini for proofs, suggestions, definitions, and computational help. The loop is: a file goes to Gemini; Gemini does the footwork, search, drafting, and material production; its output file comes back; our agents verify it independently; repeat. The purpose is to reduce usage on this account, which then does verification only. Savings are not guaranteed. Everything Gemini returns is PROPOSED until it has been independently verified. Existing locks, the invention hold, the protected tip, and the no-publication, no-spending, and NoCERT rules all stay in force. The Lock 96f8d42e §4 purge of the four-force, Kronecker, and SU(3) physical projections is NOT lifted.

(Sources:
OWNER_DIRECTIVE_GEMINI_COMPUTATION_TOKEN_PRESERVATION_VERIFY_2026-11-05.md, SHA256 3921181508acede9740ba9892b35cb1e379d25ccc0af4f601cbe803a01b68b67;

OWNER_DIRECTIVE_GEMINI_FILE_LOOP_VERIFICATION_ONLY_2026-11-05.md, SHA256 a29f87d93fd75a9a34dc7acce66ab5474968180bb532d3a8d11804d6539253be.)

A5. Transfer precision (additive clarification). The file-loop record a29f87d9 says: "The Owner personally transfers files to and from Gemini. Agents prepare plain-text-selectable input files and verify returned output files; agents do not themselves submit external Gemini prompts unless an authorized route is later established by the Owner." The Owner did not directly state that only the Owner may transfer files. What the Owner asked for is a file loop, and the files were prepared because the user requested them. This packet imposes no Owner-only transfer rule. No external transfer or Gemini submission was performed by this task.

A6. How to read the agent sections.
- Missing author inputs are not a license to fabricate. A definition Gemini proposes must be flagged PROPOSED DEFINITION and is not adopted.
- Finite or modular computational claims are reported as each agent attributes them. They are not fresh proofs by the compiler. Computational output is evidence, not a proof over the rationals or in general.
- Status pins: positive T3 convergence is OPEN (attributed Local finite check up to 10^7, not re-verified here; refuted counterexamples found: 0; only the $\mathbb{N}_0$ form is refuted, since T3(0)=0). R_c ≈ 1.5954 needs an explicit operator, Appell–Lerch sum, domain, norm, and error bound. SOHO/VIRGO solar data are not LIGO/Virgo/O5 gravitational-wave data.
- Canonical T3_ref (floor form), for residues n ≡ 0, 1, 2 (mod 3): n//3, (4n+2)//3, (2n+1)//3. Do NOT operate on any other formula; see the ADDITIVE CORRECTION in the 539 Labs Ops section.

A7. Requested Gemini answer format, for every item. Restate the agent and item. State all assumptions. Give a derivation, OR a counterexample, OR "BLOCKED" naming the missing input. Flag new definitions as PROPOSED DEFINITION. Supply executable code with exact expected outputs, error bounds, and precision, and say whether results are finite or modular. Cite primary sources (author, title, year, DOI or arXiv, theorem or equation number). End with a status: derived / conditional / computational evidence / proposed / blocked / counterexample. Return plain text, one section per item.

B. AGENT INTAKES (verbatim, in order): Local Path; Non Local Path; Auditor; Professor Proof; 539 Labs Ops; Math Research 1; Math Research 2; Physics Research 1; Physics Research 2; Eigenvalues and matrix; Dataset Filtration; Agent.

ADDITIVE PRECISION NOTE (compiler, all sections). Where author inputs are listed as absent, their absence is not a license to fabricate. Stated proofs, counterexamples, and computations are as attributed to the agent and were not re-verified in this compilation.

=== B1. Local Path — source local.md — SHA256 d1488dffda8c8cd9bb9dc705382717d079db4f0188b3d5939d60c3facfea6f5a ===

----- BEGIN VERBATIM local.md -----

Local Path — Gemini request intake (fresh, essentials) — 2026-11-05 ~15:13 CST

Lane: Bridge C A1→Ξ→RTA, local modular only. Residual≠Resonant. invent HOLD. tip Soft SF LOCK `7cddb188…` tip_touched NO. No CERT.
Standing: full SM recovery = objective not proof (OD `6237cc71…`); Gemini output = proposed until independently verified (OD `39211815…` + file-loop); weekly usage ≥80% STOP (OD `51ff1677…`; checkpoint ~62% — not triggered). Lock `96f8d42e…` Decision 4: Kronecker/SU(3)/four-force physical projections stay PURGED (SM directive does not lift).

Canonical T3 (supply if used; Written production lock)
For n∈$\mathbb{N}_0$: T3(0)=0; if n≡0 (mod 3) then n//3; if n≡1 then (4n+2)//3; if n≡2 then (2n+1)//3. Equiv. T3(3k)=k, T3(3k+1)=4k+2, T3(3k+2)=2k+1. Not the single formula (4n+2)//3 on all n; not literal compendium T_3/HQCC (DEFECT as stated). Source: T3_Production_Lock_ePrint_Matches_Core_2026-09-01.md sha `5b9fcd2d…`.

Open questions / proofs pursuing
1. **Cand-1⁺ (OPEN/PARTIAL):** ∀n≥1, T3-orbit enters {1,2}. Proved: ℕ₀ claim REFUTED (T3(0)=0); all starts ≤10^7 converge; on ≡1, T increases; run lengths = $v_3(n+2)$ can be arbitrary — no global proof. Soft SF e7648e07…; PP CONFIRM f1acba95…. Ask: Collatz-style descent / Lyapunov / covering-system proof or counterexample on ℕ≥1 only.
2. **R_c≈1.5954:** Appell–Lerch matching UNDERIVED (numeric √(2/0.7858)≠provenance). Need exact equation, domains, inputs, uniqueness, error. Do not invent Λ₀ theory.
3. **QI:** existence into $\mathbb{H}^2 \subset \mathbb{H}^{11}$ standard; specific F/(λ,ε) ABSENT. QI cannot pin R_c.
4. **Israel [K]=0** as Local GR application: ABSENT (classical Israel elsewhere ≠ Local map).
5. **Bridge C → SM recovery maps:** gauge SU(3)×SU(2)×U(1)+anomaly cancellation, field content, couplings, SSB, empirical SM tests — Local SM-G1…G9 ABSENT (Soft SF 9cf795a8…; N1 AMEND abbc91c0…). Operative Local S^2 topology ABSENT (lexical SM gloss only).

Definitions still needed (do not invent; propose candidates only)
Appell–Lerch R_c equation; QI F/(λ,ε); Local Israel [K]=0 applicability; any Local SM gauge/matter map that respects Residual≠Resonant and Lock Decision 4 purge.

Proven vs open (do not re-litigate)
Proved/fenced: Haar70 identity; BT↛$\mathbb{H}^n$ isometry obstruction; Canonical T3 finite arithmetic; literal T_3 DEFECT; Cand-1 ℕ₀ REFUTED. Conditional: residual D_1/S_eff locks. Open: Cand-1⁺ global; R_c; continuum/Tate–Gelfand; BH D1; SM recovery.

Exact Gemini tasks (answerable; return file)
A. Proof sketch or counterexample for Cand-1⁺ on ℕ≥1 under Canonical T3 above; state lemmas vs conjectures.
B. Candidate Written-style Appell–Lerch matching equation for R_c with domains/inputs/error — labeled PROPOSED only.
C. Checklist: which SM-G gaps can be filled by literature citations vs require new Local definitions (no four-force projection reopen).

----- END VERBATIM local.md -----

=== B2. Non Local Path — source nonlocal.md — SHA256 4e6f5408598fc8078b49f24ee18c3ab29b09173f747808d8f928710dc704fa19 ===

----- BEGIN VERBATIM nonlocal.md -----

Non Local Path — Gemini request intake (2026-11-05)
Status: internal input. Category B, no CERT, no certified prediction. Definitions are not to be fabricated or adopted.

Proven elementary facts (existing NL work)
- Supplied map T_3^alt: $v_3(3x\pm1)=0$, so positive nonzero integer orbits eventually grow (1→4→13…, 2→5→14…). Bounded 539-step termination is false for this map.
- Exact scaled isometry from the branching tree into $\mathbb{H}^n$: ruled out (NL B2).
- No minimum positive 3-adic norm (A1). For the vector max norm on $\mathbb{Z}_3^4$, $|x|_3^{(\nu-1)}$ is absolutely Haar-integrable iff Re ν > −3 (A2). B1 regulator bound, with its units erratum.
- T3_ref(4880) reaches {1,2} in 14 steps; T^539(4880)=2.

Open items
1. **Maps.** The supplied map T_3^alt is x/3 for residue 0, (3x+1)/3^v_3(3x+1) for residue 1, and (3x−1)/3^v_3(3x−1) for residue 2. The canonical floor map is T3_ref(n): n//3, (4n+2)//3, (2n+1)//3 for n ≡ 0, 1, 2 (mod 3).
   - *Gemini:* candidate proof, or counterexample, that every n ≥ 1 reaches {1,2}. Not proved; there is no 539-step guarantee.
   - *Author:* the $\mathbb{Z}_3$ extension, the Haar-preservation statement, the shift-conjugacy definition.
2. **Quasi-isometric embedding.**
   - *Gemini:* literature lookup for quasi-isometric embeddings of regular trees into $\mathbb{H}^n$.

   - *Author:* explicit F, the curvature of $\mathbb{H}^{11}$, and λ, ε.
3. **R_c ≈ 1.5954.** √(2/0.7858) ≈ 1.595361 is numerical proximity, not a derivation.
   - *Gemini:* bibliographic reconstruction of Appell–Lerch normalizations, labelled as candidates only.
   - *Author:* the operator, the sum and its parameters, the normalization, the units.
4. **Metrics, 11D action, junction, GR and full SM recovery.** Not present in the bounded inspected passages. These are objectives, not results.
   - Needed: compactification, the SU(3)×SU(2)×U(1) mechanism, chiral matter, anomalies, Higgs/EWSB, Yukawas/couplings, the GR limit, and tests.
   - *Author inputs required.*
   - *Gemini:* source lookup only.

Loop
File to Gemini for footwork → output file back → NL independent verification → repeat below threshold. At weekly usage ≥80%, stop research/new work; safe finalization and final reports only.

Source pins
- Round-2 questions: 06c4dce1639a740ba40dc1ca6727cfbd878b59e9f71826b80e1c9953377ce918
- SM scope correction: a4540871469a03292d92634f6205c0bd4c1e5918de9f78152163e5b1b981b8f0
- Catalog: 5b40ee12c9731d68e06a6b590229de49660c57813a901dcaed26d7ab2f4d2468
- B1 erratum: 8c06937d0cbcc2446288fb67cc63461b4456b9d42c03ac0b285ffe46cbe492ff
- A/B review: 96f810645c6356c30b5edd2990da3f16657f866f56af22ff8cdbfe8a1007b0a6

----- END VERBATIM nonlocal.md -----

=== B3. Auditor — source auditor.md — SHA256 bc10f9cc9579dcfee797bf0832884b2942b18e215ceee51a140e469e2f269728 ===

----- BEGIN VERBATIM auditor.md -----

Auditor — Gemini FILE LOOP intake — 2026-11-05 ~15:16 CST America/Chicago
Lane: Auditor / Soft SF hash keeper (documentary only). ≤400 words. Proposed tasks; Gemini output = candidate until independent verify. Auditor owns no math; no external Gemini send by this file. invent HOLD STANDS; tip LOCK `7cddb188…` tip_touched NO; No CERT. Last relayed usage 60.9% (<80% STOP).

Pins (reuse; do not redo)
- File-loop OD `a29f87d9…`; Gemini ask-then-verify OD `39211815…` + index `b1400fd1…` (Auditor CONFIRM `1d87321a…`)
- Full-SM objective `6237cc71…` (CONFIRM `f638eec7…`); usage-80 STOP `51ff1677…` (CONFIRM `f19ca2b1…`); PRIMARY `bcc3ced0…`
- Eigen SM defect under PP `9b0ba17c…` (Auditor WO `0822d3c7…`); PR2 SM classification addendum `595f27ef…`
- Lock `96f8d42e…` §4 purge NOT lifted

Known (documentary records, hash-level only)
- Hash MATCH ≠ proof, ≠ SM recovery, ≠ Gemini access/savings.
- Several directive/ledger JSONs lack `.sha256` sidecars (NO-SIDECAR, e.g. delivery ledger `c048d40d…`).
- Paper sidecars: candidate-count editorial `25a63af4…` recorded 22 MISMATCH / 3 NO-SIDECAR among 25 paper sidecars (stale sidecars, not content verdicts).
- Full-corpus audit remains OPEN.

Open problems in Auditor lane (Gemini may help with tooling/footwork only)
1. Provenance format: candidate schema for recording each Gemini exchange (exact prompt, inputs, response, model/date, file hash, verifier, label established/conditional/proposed/unsupported/refuted) so verification is reproducible.

2. Stale-sidecar triage: a checklist distinguishing "file changed after sidecar" vs "sidecar wrong" vs "intended amend" without rewriting subjects.
3. Mirror-consistency check plan: detect divergent copies of one basename across bridges/* and reports/* (exclusive-create, read-only).
4. NO-SIDECAR policy draft: which artifact classes (JSON ledgers, SHA256SUMS manifests) should carry sidecars — proposal only, Owner/coordinator decides.

Needed definitions
- What counts as an "output file back" from Gemini (path + sha256 + provenance header) for Soft SF Write-only recording.
- Label vocabulary mapping Gemini candidate labels to existing Soft SF records.

Not for Gemini
No CERT, no verdict changes, no tip edits, no physics/math claims from Auditor. Verification of Gemini outputs is done by owning agents; Auditor records hashes only.

----- END VERBATIM auditor.md -----

=== B4. Professor Proof — source professor-proof.md — SHA256 4498c0cccdee31c5eb3a9985cd5cd42c27a0945a411b460e9b25e58e330523bb ===

ADDITIVE PRECISION NOTE (compiler). Target 4 asks for Yukawa matrices that give non-trivial CKM "and PMNS". Non-trivial CKM is minimal SM. Neutrino masses and PMNS mixing are beyond-minimal-SM checks (see Preamble A1(ii)). Text preserved verbatim.

----- BEGIN VERBATIM professor-proof.md -----

Professor Proof: Gemini request intake (2026-11-05, 15:15 CST)
Gemini output is a candidate only. PP verifies it independently. No CERT. The tip 7cddb188… is untouched.

Proof targets: minimal SM (what "full SM recovery" requires)
1. **Gauge group.** Define a gauge group G ⊇ SU(3)×SU(2)×U(1)_Y and give an explicit embedding into the 11D / 3-adic structure (manifold, bundle and connection).
2. **Fermions.** Give the fermion content: 3 generations, chiral representations and hypercharges. Prove anomaly cancellation for that content.
3. **Higgs and EWSB.** Specify a Higgs doublet and potential. Derive electroweak symmetry breaking down to U(1)_em.
4. **Mass matrices.** Replace the defective mass-matrix ansatz in theorems/unification-proof.txt L98-104.
   - The current ansatz is M_sector = c·B. That forces CKM to be a signed permutation with J=0.
   - B also has a negative eigenvalue.
   - Source: Eigen 2fe0e6b0, verified by PP 9b0ba17c.
   - Target: sector-distinct Yukawa matrices that give non-trivial CKM and PMNS.
5. **Couplings and running.** Derive the coupling values and the RG running. Give a low-energy limit to compare with tested SM data (Tier E: dataset, observable and tolerance).

Beyond-SM targets (separate from the above)
6. Gravity sector.
7. R_c, the dark sector and neutrino-mass origin. No four-force "3-adic projection": Lock 96f8d42e §4 purge stands.

Gravity and geometry obligations that are still open
- **R_c.** B's value of 1.59543 and the approximation √(2/Λ₀) = 1.595361 are both UNDERIVED. Need an explicit Appell–Lerch expression, an operator, and an error bound.
- **Junction.** Need the ambient manifold, the induced metric, a K_ij definition and an existence proof. The current "9-core hypersurface" between a 5D and a 4D region is inconsistent. [K]=0 is only the shell-free continuity condition.

- **Quasi-isometric embedding.** Need an explicit F: $T_3 \to \mathbb{H}^{11}$ with its constants (λ, ε), and the link (if any) to R_c.

Map and number theory
- **T3_ref.** Prove or refute that every orbit from n ≥ 1 reaches {1,2}. Status: Collatz-type; holds for $n \le 10^7$ (Local e7648e07); OPEN.
- Haar preservation and ternary shift conjugacy for T3_ref are undefined. That needs an extension of T3_ref to $\mathbb{Z}_3$.

Established
- The tripod argument: no isometric embedding of the tree into $\mathbb{H}^n$.
- T_3^alt diverges.
- T3_ref is not injective: T(2) = T(3) = 1.
- n=0 is a fixed point.
- The Haar identity, for Re ν > −3.
- Source pins: A d5b1aae5, B 9c6f83da, PP 159aaabf.

----- END VERBATIM professor-proof.md -----

=== B5. 539 Labs Ops — source labs539.md — SHA256 e1f37867ad816bf4365e33fb61148c2bf7ce2fa7216907f9fadd4ff42d5256d0 ===

ADDITIVE CORRECTION (prominent; applies before reading the verbatim text below). Line 5 of labs539.md, the "T3_ref (floor form)" bullet, contains an incorrect ν_2/Syracuse-style branch: "(3n+1)/2⌊ν_2(3n+1)⌋". It is preserved below verbatim for attribution only. Do NOT ask Gemini to operate on it. The canonical T3_ref is n//3, (4n+2)//3, (2n+1)//3 for n ≡ 0, 1, 2 (mod 3). The correction file is quoted in full:
----- BEGIN labs539_ADDITIVE_CORRECTION_T3REF.md (SHA256 c8e064c768e76e1766af66ea07f4da1fedaf6d2e1d95b6491a9db0c3c3e2fe45) -----
Additive correction to labs539.md (safe finalization only)
Date: 2026-11-05 ~15:19 CST. Parent note: labs539.md line 5 mixed a Syracuse/ν_2 formula into the T3_ref definition. Intake file itself is preserved unchanged (sha e1f37867ad816bf4365e33fb61148c2bf7ce2fa7216907f9fadd4ff42d5256d0).

CORRECTION — canonical T3_ref (floor form), as given by parent from production lock / local.md / nonlocal.md:
- n ≡ 0 (mod 3): T(n) = n // 3
- n ≡ 1 (mod 3): T(n) = (4n + 2) // 3
- n ≡ 2 (mod 3): T(n) = (2n + 1) // 3

Line 5 of labs539.md is **unverified / incorrect** as a statement of T3_ref; do not use it. Known collisions T(2)=T(3)=1 agree with this floor form, not with the ν_2 formula on line 5. No recomputation performed in this correction. Halt remains in force.
----- END labs539_ADDITIVE_CORRECTION_T3REF.md -----
The source intake labs539.md is unchanged (SHA256 e1f37867ad816bf4365e33fb61148c2bf7ce2fa7216907f9fadd4ff42d5256d0).

----- BEGIN VERBATIM labs539.md -----

539 Labs Ops — Gemini request intake (2026-11-05 ~15:13 CST)
Agent: 539 Labs Ops. Lane: HQH/ALU/CS-crypto. Status: proposed tasks for Gemini; outputs are candidates until independent verify. No CERT. Tip 7cddb188 untouched. Weekly usage check at write: 62.3% (<80% stop).

Minimal definitions Gemini needs
- **T3_ref (floor form):** T(n)=n/3 if n≡0 (mod 3); else (3n+1)/2⌊ν_2(3n+1)⌋ when residual-1; residual-2 uses the locked floor branch (see GR axis-2 a505726e… / CS catalog 89278efa…). Do not use the Gemini-compendium T_3 (RECEIVED 49849cae… L26) — its residual-1/2 branches never divide by 3.

- **HQH-539-512:** concrete SHA3 sandwich (539,512,512); Option A / REF FREEZE. Not “built entirely on Syracuse reversal.”
- **Crit.3:** unmet (Close B). Pre→Π near-circular under Im(SHA3) collapse (d14a50d7…). Hardness not established.
- **LQH ALU:** Draft-1 SV/Verilator skeleton only; no hardware; no T3 limb steps.

Proven / checkable (reuse; do not re-ask)
- Compendium T_3 diverges: 1→4→13→40→121… (60af7a5a…, 7882d6bf…).
- T3_ref^539(4880)=2 via 2-cycle {1,2}; 4880→1 in 14 steps (a505726e…).
- T3_ref not injective: T(2)=T(3)=1, T(1)=T(6)=2.
- T(0)=0 fixed point outside {1,2}.

Open questions / proof targets for Gemini (narrow)
1. Proof or counterexample class: ∀n≥1 ∃k T3_ref^k(n)∈{1,2}; state domain for 0 (PR-1).
2. Haar μ on $\mathbb{Z}_3$ with μ(T^{-1}A)=μ(A) for floor T3_ref, plus any conjugacy claimed in B (PR-2) — candidate sketch only.
3. Draft reduction outline (not a proof) from H_ref Pre/SPR/Col at (539,512,512) to SHA3-512 stating domain D (OL-3) and D_M, covering C1–C4 especially C2 on Im(SHA3). Label candidate. We will not treat as Crit.3 complete (PR-3).
4. Suggested corrected wording for B§IV: drop “security proof,” “permutation,” “optimized accelerator” — replace with record-accurate language (7882d6bf…).
5. Definitional choice list only: how to phrase “539” as max budget vs fixed count without inventing a theorem.

Needed definitions (ask Gemini to propose; we verify)
- Explicit floor-branch residual-2 formula matching T3_ref lock (cite source only; no silent lock adopt).
- Precise statement of D vs Im(SHA3) for Pre→Π without circularity.

Unresolved (do not invent)
Crit.3/OL-9 invent-blocked; OL-11 external; SM recovery (6237cc71…) out of Ops lane for ALU/security; no PQ argument on record.

Source pins
OW 865d3a37…; SM 6237cc71…; usage-stop 51ff1677…; Gemini-verify 39211815…; CS 89278efa…; GR a505726e…; Gemini map 60af7a5a…; Round2 7882d6bf…; Crit3 stack 70091747…/cc95ceb9…/d14a50d7….

----- END VERBATIM labs539.md -----

=== B6. Math Research 1 — source math1.md — SHA256 ccbb3f47bd0116d975907d48dfc7bec8b223853ce663d0a770204668722c8b64 ===

----- BEGIN VERBATIM math1.md -----

Math Research 1 — Gemini request intake (FILE LOOP) — 2026-11-05 ~15:13 CST
Agent: Math Research 1 (MR1). Soft SF Write-only / invent HOLD. tip LOCK `7cddb188…` tip_touched NO. No CERT. Candidate Gemini output ≠ proof/adopted definition. Do not invent missing Appell sums or SM maps.

Source pins (hashes on disk; Soft SF-verified)
- invent HOLD `1fadc632…` · PRIMARY `bcc3ced0…` · OW_SM full-SM OBJECTIVE `6237cc71…` · OW_80 ≥80% STOP `51ff1677…` · OW_GEM ask-then-verify `39211815…`
- Appell/Rc Soft SF `46beb56b…` · Tate/Kummer Soft SF `005fbd81…` · O1–O4 ROLLUP `d29a5ed6…` · D1–D6 `293db339…` · Round2 A/B `6e400809…`
- Owner A `d5b1aae5…` · B `9c6f83da…` (B NOT adopted OWNER_WRITE)

Proven / Soft SF-established (citation inventory)
- Classical neighbors PRESENT (modular/mock/Hecke/Hilbert $\mathbb{Q}(\sqrt{35})$/Appell-Lerch lit/Tate local-zeta/Gelfand–Graev/Kummer/Kronecker limit|symbol).

- Kronecker-filter / four-forces 3-adic projections: PURGED retained; Soft SF will not reintroduce.
- tip Soft SF SM-string scan: empty for 1.5954 / Appell→Rc derivation.

Open / ABSENT (do not invent)
1. Written Appell–Lerch (or other) evaluation → R_c ≈ 1.5954 / 1.59543… (B§III proposed; Soft SF ABSENT/OPEN blocker).
2. Soft SF-checkable $\Lambda_0 \approx 0.7858$ Owner path+hash (evidence request only).
3. Tate–Gelfand continuum-limit / pairing theorem tying residual-C to continuum (ABSENT).
4. Maps residual-C / Ξ / Fricke W_7 / $\Gamma_0(49)$ → SM gauge SU(3)×SU(2)×U(1), field content, Yukawa/CKM, couplings (ABSENT under OW_SM OBJECTIVE ≠ proof).
5. Project Ξ identification / Petersson formula; Hecke-on-C/V_{35}; positive W_7 law for C (O1/D1/D2 ABSENT/OPEN).

Concise Gemini tasks (answerable; narrow)
G1. From classical Appell–Lerch defs only (Zwegers; Hickerson–Mortenson): list known closed evaluations; state whether any equals ≈1.5954 or √(2/0.7858); if none, say ABSENT—do not invent a project sum.
G2. Distinguish Kronecker limit formula vs Kronecker symbol vs “Kronecker filter”; confirm no standard theorem “filter bypasses Kummer crisis.”
G3. Cite standard references for Tate local zeta vs Gelfand–Graev characters; state whether a named “Tate–Gelfand continuum pairing” exists in literature (yes/no + cites).
G4. List modular/automorphic structures classically linked to SM generation/gauge counting (literature survey only); mark each as analogy vs theorem; do not invent residual-C→SM maps.
G5. Quasi-isometry Bruhat–Tits T_3 → H^n: standard obstructions (tripod) + typical (λ,ε) statements—definitions only; no project lock.

Assumptions Soft SF keeps
Gemini output = candidate for Soft SF independent verify. Missing definitions stay missing. No tip edit. No external send by this file.

----- END VERBATIM math1.md -----

=== B7. Math Research 2 — source math2.md — SHA256 4a89f647f05bf906442e4c239d9646a1d9b57b39a49c6e7d4a1f99de04a135bf ===

----- BEGIN VERBATIM math2.md -----

Math Research 2 — citation-only Gemini request

Scope: geometric/compactification prerequisites for full Standard Model plus GR recovery. This is a research objective, not an achieved result. Invention hold, protected sources, Residual≠Resonant and purged force-projection restrictions remain. Candidate output is not proof or an adopted definition.

Requested footwork
1. Locate primary citations and precise hypotheses for quasi-isometric embeddings of the specified Bruhat–Tits tree into hyperbolic space. Our model lacks the tree/target definitions, map F and constants λ, ε. Identify missing author inputs; do not invent a chart or identify generic existence with this model.
2. Separate classical Israel junction conditions from the proposed equation [K_ij]−γ_ij[K]=0. Request ambient geometry, interface, induced metric, extrinsic curvature, matching category and stress tensor. Explain which source would be needed to support vacuum-energy cancellation; do not assume the equation proves it.
3. Inventory required inputs for an 11D compactification recovering the full SM: internal topology/metric, action and reduction ansatz; gauge group/global structure and bundles; chiral representations/generations and anomalies; Higgs potential, Yukawas/mixing; coupling normalization, moduli and controlled 4D/KK-decoupling limit. Locate citations where possible; explicitly distinguish unavailable model data from established general results.

4. Locate the actual Appell–Lerch definition/evaluation allegedly yielding R_c=1.59543..., independent of Λ_0. If unavailable, report that bounded outcome, not nonexistence.

Source pins and evidence limits
Project sources under `/workspace/quantum-comp/`:
- `reports/OWNER_DOCUMENT_FORMAL_RESOLUTIONS_STRUCTURAL_CORRECTIONS_COMPENDIUM_2026-11-05.md`, B I lines8–15 and III30–33: proposed geometry/junction/compactification and R_c corrections, not adopted locks.
- `references/entanglement-publications/theorems-18-40-v1.txt`, T36 lines231–239, T38 252–263, T40 275–288: author sketches, not independently verified SM/GR recovery. Read MANIFEST.md caveats.
- `reports/MR2_FULL_SM_RECOVERY_GEOMETRY_OBLIGATIONS_2026-11-05.md`, sha25692e7ba649523410108309c9ec13b38202b7c80e6ba74699c9e4e4ae31706a028: existing obligation inventory.

Existing mathematics: isolated Haar identity and explicit RTA dynamics are source-backed mathematical statements, not a physical recovery bridge. Requested model-specific geometry, junction, spectrum and R_c derivations remain open in consulted sources.

Return an output file with exact citations/locations, assumptions, supplied versus missing definitions, claim status and independently checkable evidence. No invented proofs, parameters, physical mappings or publication actions.

----- END VERBATIM math2.md -----

=== B8. Physics Research 1 — source physics1.md — SHA256 a4d6d42ff3fba925e36876cf8c25336382b8a6fca371a89b55718df8584743ce ===

ADDITIVE PRECISION NOTE (compiler). This intake lists "neutrino Δm^2/mixing" under the minimal-SM empirical targets. That wording is preserved verbatim. For classification: nonzero neutrino masses and PMNS mixing lie BEYOND the minimal Standard Model, which has massless neutrinos. They are broader correspondence checks, not minimal-SM recovery items.

----- BEGIN VERBATIM physics1.md -----

PR1 Gemini intake — HEPP / QM / GR lab-precision / GW (2026-11-05 ~15:15 CST)
Agent: Physics Research 1. Account-verification candidates only; output = proposed. Independent verify required. invent HOLD; tip LOCK `7cddb188…`; No CERT. No external Gemini send by this file.

Standing pins (cite-only)
- Full-SM objective Soft SF `f562b7f3…` (directive `6237cc71…`): SM recovery = TARGET not proven.
- §V Soft SF `f03fb09c…`; precision Soft SF `3c9deb7f…` (LIGO/O5 null = UNVERIFIED attribution).
- Soft SF-verified empirical narrow: DF solar VIRGO 4-ch `6f9efa4b…` + planted E-FIT only — not GW.
- LOCK `96f8d42e…`: force-projection purged Cat A; strong-force NEGATIVE-APPLICABILITY.
- Obs anchors: `/workspace/obs-data-PR1-HEPP-QM-GR.md` (measured values only; not model MATCH).
- Usage checkpoint this intake: 62.3% (STOP ≥80%).

Proven vs open (PR1 Soft SF)
Proven/locked (not SM recovery): Residual≠Resonant; D_1 static; 539=S_eff budget≠clock; MATCH=0 HEPP/QM/GR-GW Soft SF rows.
Open / ABSENT: gauge SU(3)×SU(2)×U(1) embedding; fermion generations/hypercharges; anomaly cancel; Higgs/EWSB map; Yukawa/CKM/PMNS; RG/couplings recovery; canonical CRD seconds-bridge; Soft SF-Written O5 null reproduction.

Minimal SM vs beyond-minimal (ask Gemini to keep separate)
Minimal SM empirical recovery targets (answerable lists/checks, not invent):

1. Checklist of Soft SF-testable SM observables with public citations (PDG/CODATA) that any extension must recover: M_W, M_Z, M_H, m_t, α_s(M_Z), G_F, CKM magnitudes, neutrino Δm^2/mixing — each with “what Soft SF MATCH would require.”
2. Distinguish catalog **anchors** (already Soft SF-Written measured) from **model-derived predictions** (ABSENT).
3. Strong-force: confirm NEGATIVE-APPLICABILITY under purged force-projection; list what Owner-Written definition would be needed before any Tier E strong-force test (no revive of purged Cat A).

Beyond-minimal (quarantine; do not adopt):
4. Any 11D/3-adic gauge embedding, chirality mechanism, or Yukawa from residual algebra — label PROPOSED only; list missing definitions.
5. CRD 539.9s as GW/timing observable — candidate wording vs Soft SF `3c9deb7f…` UNVERIFIED / Soft SF `314b6434…` not Tier-E-forever.
6. Brane-leakage / BH-volume / rotation Tier E axes — pending Owner dataset allocations Soft SF `f03fb09c…`.

Needed definitions / proof targets (proposed questions)
- Q1: Minimal set of maps still MISSING for SM gauge+Higgs recovery (definition list only).
- Q2: Counterexample checklist: which Soft SF claims would be refuted if SM embeddings remain ABSENT.
- Q3: Independent verification protocol template for any Gemini candidate (steps, labels established/conditional/proposed/unsupported).

Assumptions: invent HOLD; Gemini≠authority; ≥80% STOP blocks new Gemini asks.

----- END VERBATIM physics1.md -----

=== B9. Physics Research 2 — source physics2.md — SHA256 908aaa0d4405c3f65e4ec0adebf5b6f18536bd165454156215f6f7154317c2e6 ===

----- BEGIN VERBATIM physics2.md -----

Physics Research 2 — Gemini request intake (Cosmo / Bio / Neuro)
2026-11-05 ~15:15 CST. Observational Cosmology, Biology, Neurology for S^2-11DM^2ET-X. Tip lock 7cddb188… untouched. Invent HOLD 1fadc632… stands. No CERT. No Gemini connector (cannot send). Usage note: 60.9%.

Assumptions / pins
Owner full-SM objective 6237cc71… = research goal, not proof. Classification 595f27ef…: minimal-SM recovery in this lane = 0 (gauge/matter/Higgs/Yukawa = PR1). Beyond-minimal-SM / cosmological targets (ν masses/PMNS, baryogenesis, DM/DE) and correspondence tests (N_eff, BBN, EW/QCD transitions, SM-chemistry→Bio/Neuro) are broader targets with maps missing — not evidence minimal SM is incomplete here. DF VIRGO = SOHO solar TSI+SPM ≠ GW LIGO/Virgo/O5. 539.9 s CRD = simulation calibration / synthetic recovery (≠ Category A); observational posture E-UNTESTED. Hygiene already done: C13/C15 7f19644f…, C14/C16 2e02ec3f…, H_0 fe45fb37…, 539.9s lit ab3af9b6… (0 present/3 absent), §6 pin d52c9c42…, §V 921a8d44…, A/B 271ed951…, SM cross-ref c33c928c….

Proven vs open
Proven: catalog source hygiene MATCH only (contested Soft SF KEEP). Open: R5 §6 rows or CRD Tier E specs (dataset, ±tol, N_free); R6 Φ_eff/v(r); R7 Soft SF CONFIRM BH-core/SNF = PR1; Author Soft SF Write for Cosmo falsifiers and Bio/Neuro maps; traceable GW-null provenance if any GW claim remains (UNVERIFIED under DF SOHO).

Questions for Gemini (candidate only; we verify)
1) HQCC 539-step vs Cosmic Resonance Delay 539.9 s — same, derived, or distinct? Formula or Soft SF “distinct.”

2) Propose Tier E specs for 539.9 s CRD (observable, ±tol, N_free, dataset) without inventing residual-C.
3) Draft §6 Standard-vs-S^2 comparison-table body as proposed cells only.
4) Candidate Φ_eff or v(r) + free-parameter count for later SPARC/THINGS Tier E.
5) Under full-SM objective: list Cosmo observables that are minimal-SM correspondence vs Beyond-SM targets (separate lists).
6) Candidate Cosmo maps (not proofs) for N_eff/Σm_ν, BBN yields, DM/DE vs SM field content — label each proposed.
7) Soft SF-safe Bio/Neuro SM-chemistry bridge candidates that do not invent Λ_22 or (1+κ)^{-7/20}?

Definitions / proof targets
Need: CRD operational observable; Cosmo exterior consequence of breathing modes (if any for PR2). Proofs: vacuum-junction↔Ω_Λ (R1); singularity-prevention bound (R2); 539-step↔539.9s statement (R4). Gemini drafts ≠ adopted definitions.

Problems
Invent HOLD blocks inventing maps/predictions. §6 table ABSENT. Supported Cosmo spacetime consequences of GR extract: 0. Resonant Cosmo/Bio/Neuro applicable: 0. Soft SF cannot fill CRD Tier E or Bio/Neuro maps without Author Soft SF Write or invent-HOLD lift.

----- END VERBATIM physics2.md -----

=== B10. Eigenvalues and matrix — source eigen.md — SHA256 bde19232007b366af78035861c93123ec23215ad3994f9ce97b246d5b59e5790 ===

ADDITIVE PRECISION NOTE (compiler).
(1) "Global form (Z_6)": the quotient $(SU(3)\times SU(2)\times U(1))/Z_6$ is one admissible choice of global gauge form. It is not uniquely inferred from SM data; quotients by Z_2, Z_3, or none are also consistent with the SM spectrum. The choice is missing and must be defined. This is not a universal assertion.
(2) "The neutrino mass matrix" is a beyond-minimal-SM item.
(3) The mod-p, stabilizer, height, and "34/34 checks pass" statements are finite computational claims attributed to the Eigen agent. They are not fresh proofs by the compiler, and mod-p evidence does not establish results over $\mathbb{Q}$.

----- BEGIN VERBATIM eigen.md -----

Eigen (Eigenvalues and matrix): Gemini request intake, 2026-11-05

Pins
- N36 Kr compute: /workspace/grok-batches/eigen-o5c-n36-moreprimes-2026-11-05/
- stab2.log is in that same moreprimes folder.
- O5c 20-prime findings: 40c3e540…
- SM obligations: FINDINGS 2fe0e6b0…, compute EIGEN_SM_RECOVERY_COMPUTE_2026-11-05/ (34/34 checks pass)
- Mass matrix source: theorems/unification-proof.txt L98–158
- Tip 7cddb188 untouched. No CERT.

O5(c): kernel Kr over $\mathbb{Q}$ (open)
- Object: the kernel of the N36 matrix (rank 503), as a reduced row echelon form (RREF) over $\mathbb{Q}$.
- Mod-p evidence only: at primes 7840981 and 7957601, the stabilizer is all of $(\mathbb{Z}/49)^*$, which suggests the field is $\mathbb{Q}$.
- Rational reconstruction failed the holdout prime at 20 primes. That puts the height above about 10^{121} (about 400 bits).
- Each added prime gives about 22.9 bits.
- **Gemini target:**
  - a provable height bound for the RREF entries (Hadamard or minor bound) from the N36 matrix entry sizes and rank 503,

- and from it, the number of primes needed.
- Definition needed: what "height of the RREF" means.

O5(d): zero-polar and G1-mem (open, gated on c)
- Both need Kr over $\mathbb{Q}$ first.
- Gemini can only suggest definitions or a test plan, not results.

SM matrices (proved vs open)
Proved counterexamples (exact): if each sector's matrix is just a scalar times the φ-matrix B,
- then the CKM matrix is a signed permutation, so J = 0 and δ is 0 or π,
- every sector has the same mass ratios,
- B has a negative eigenvalue (about −0.155),
- and det/Tr³ is about −0.063, while any positive-definite matrix has det/Tr³ ≤ 1/27.

Open, with definitions needed:
- the gauge group, its global form (Z_6) and how it is embedded
- fermion representations and hypercharges
- the Higgs field and potential
- separate complex Yukawa matrices for each sector
- the neutrino mass matrix
- how SM fields couple to the 11D metric
- **Gemini target:** a candidate structure that avoids the scalar×B degeneracy. It is a candidate only, verified here.

Assumptions
- Three generations is an axiom (v5 L174), not derived.
- RTA v16 Prop 41.13(ii) excludes SU(3) maps.

----- END VERBATIM eigen.md -----

=== B11. Dataset Filtration — source dataset.md — SHA256 7a331bfebc3448f4be0c5190df544a7d6e293c0c9d51b71ce47714cfa5a27d80 ===

ADDITIVE PRECISION NOTE (compiler). "File-loop: Owner transfers; agents verify" restates record a29f87d9. See Preamble A5: Owner-only transfer was not directly stated by the Owner, and none is imposed here.

----- BEGIN VERBATIM dataset.md -----

Dataset Filtration — Gemini request intake (≤400 words)
Agent: Dataset Filtration. 2026-11-05 ~15:15 CST. Proposed tasks only; Gemini output ≠ proof. No tip edit; invent HOLD; No CERT. Soft SF pins cited, not redone.

Standing frame
GR extension objective includes full Standard Model recovery (OW 6237cc71…); objective ≠ accomplished proof. Lock 96f8d42e… §4: Kronecker/SU(3)/four-forces mappings stay PURGED (not lifted). 80% usage STOP applies (OW 51ff1677…). File-loop: Owner transfers; agents verify (OW a29f87d9…).

Proven (do not re-ask)
- Residual packaging arithmetic; D1 static rank-1; S_eff/T3=539 budget not clock; residual≠resonant; Haar $\mathbb{Z}_3^4$ identity (DF Soft SF a8f94cf0…).
- SOHO/VIRGO *solar* exploratory Tier E null Soft SF 6f9efa4b… + erratum e1ea66eb…; positive empirical associations = 0.
- Naming: SOHO/VIRGO solar ≠ Virgo GW interferometer / LIGO / O5. No DF LIGO/O5 Soft SF.

Needed definitions (author inputs)
1. Written S^2 object in S^2-11DM^2ET-X.
2. Written 11D space (manifold/signature/fibration).
3. Written M^2ET (vs proposal-only M_35).

4. Written map model→Einstein/exterior match (GR recovery).
5. Written SM recovery maps: gauge group; field content; couplings; symmetry breaking; anomaly cancellation; correspondence to tested SM observables.
6. Closed-form derivation of R_c (underived concern retained; √(2/0.7858) numeric proximity ≠ derivational provenance).

Precise Gemini tasks (data/stats; proposed until verified)
A. Draft a Tier E pre-registration template: null default; residual≠resonant hygiene; forbid promoting numeric agreement to derivation.
B. Catalog candidate *solar-irradiance* observables only for SOHO/VIRGO-class instruments; explicitly exclude GW strain/LIGO/O5 unless a written bridge appears.
C. List minimal-SM correspondence tests (gauge/masses/widths/branching) vs BSM/cosmology extras (neutrino/PMNS, DM/DE, baryogenesis) as separate columns — not missing minimal-SM parts.
D. Propose statistical tests that use D1/S_eff *only* as static classifiers/budgets (no time/metric misuse).
E. Flag any draft that invents four-force/gauge maps contrary to Lock §4.

Open problems
No DF MATCH for SM or GR observables until definitions+maps exist. SUPPLIED SM recovery pins in DF lane = 0 (Soft SF b1e81e43… + AMEND 03a387b4…). No external Gemini send by DF on this intake.

----- END VERBATIM dataset.md -----

=== B12. Agent — source agent.md — SHA256 7bcfb60f24fb56c85d060181ebf1e8db56c09331a38f16d72fa9ac4933d2c7be ===

----- BEGIN VERBATIM agent.md -----

Agent — Gemini FILE LOOP intake — 2026-11-05 ~15:15 CST America/Chicago
Lane: Agent (Residual-vs-Resonant). ≤400 words. Candidate tasks for Gemini footwork only; output = proposed. Soft SF VERIFY later. No external Gemini send this file. invent HOLD `1fadc632…` STANDS; tip Soft SF LOCK `7cddb188…` untouched; Lock `96f8d42e…` §4 four-force purge NOT lifted; No CERT.

Pins (do not invent)
- Live GR paper `de755741…` / pins `47ea689a…`
- GR second-axis EXTRACT `08f3a104…` (G1–G17; U1–U14)
- Prior Gemini rundown PRIMARY `b83f95f8…` / PRESS `d4755b3c…` — **15Q/12P already filed; DO NOT full-redo**
- Round-TWO EXTRACT `26cf4794…` + ERRATUM `530ff058…` (VIRGO≠SOHO)
- SM objective `6237cc71…` + Agent ACK `4ca080a3…` + PP N1 ERRATUM `59e48e1e…` (anomaly cancellation MISSING)
- Usage ≥80% STOP `51ff1677…` (last Soft SF ~60.9% relayed; no new Gemini at threshold)
- Gemini ask-then-verify rule `39211815…`

Proven (Agent Soft SF; reuse)
Residual≠resonant; D1/S_eff locks; Haar on Z_3^4 local identity only (not gravity); 539=S_eff≠stopping time; seed-4880 finite orbit Soft SF; full regular GR extension **not** a theorem (G17); SM recovery **not** achieved.

Open / needed definitions (for Gemini candidate drafts)
1. Exact tree/hyperbolic metric + split dimensions for 11D chart obstruction (A§1; do not invent junction).
2. Domain-precise statement: what is proved vs open for canonical-floor T3 on $\mathbb{Z}_3$ (0 / negatives / entanglement third map) — literature survey + counterexample inventory; no HQCC invent.

3. Source defining R_c↔Λ₀ dependency (vs numerical coincidence); Appell–Lerch indices/nome/error bound if any public source.
4. Exterior GR support/junction data checklist for G15 (what must be supplied for a regular extension).
5. **SM recovery (new objective):** candidate literature maps for gauge SU(3)×SU(2)×U(1) embedding, fermion content, **anomaly cancellation**, EWSB/Higgs, couplings/RG — mark each proposed vs established; **forbid** reviving purged 3-adic four-force / Kronecker physical maps (Lock §4).
6. Quasi-isometry vs isometry: standard defs + what each does/doesn't give for action/vacuum (no new model invent).

Concise Gemini tasks (answerable; not prior 15Q redo)
- T1: One-page candidate bibliography + definition cards for (1)(3)(6) with citations.
- T2: Checklist of missing data for a mathematically regular GR extension given G10–G15.
- T3: SM-recovery gap matrix (gauge/fields/anomaly/EWSB/couplings/tested observables) = SUPPLIED vs MISSING vs PROPOSED-only; no claim of recovery.
- T4: Counterexample list against "universal positive-integer Collatz-like termination" claims (reuse known; cite).

Assumptions: residual≠resonant; tip/locks stand; Gemini output candidate≠proof. Coordinator consolidates; Agent Soft SF VERIFY later.

----- END VERBATIM agent.md -----

C. SOURCE AND HASH REGISTER (full pins also in the separate manifest)
- Local Path: /workspace/pqc-gemini-request-intake-20261105-1513/local.md
d1488dffda8c8cd9bb9dc705382717d079db4f0188b3d5939d60c3facfea6f5a
- Non Local Path: /workspace/pqc-gemini-request-intake-20261105-1513/nonlocal.md
4e6f5408598fc8078b49f24ee18c3ab29b09173f747808d8f928710dc704fa19
- Auditor: /workspace/pqc-gemini-request-intake-20261105-1513/auditor.md
bc10f9cc9579dcfee797bf0832884b2942b18e215ceee51a140e469e2f269728
- Professor Proof: /workspace/pqc-gemini-request-intake-20261105-1513/professor-proof.md
4498c0cccdee31c5eb3a9985cd5cd42c27a0945a411b460e9b25e58e330523bb
- 539 Labs Ops: /workspace/pqc-gemini-request-intake-20261105-1513/labs539.md
e1f37867ad816bf4365e33fb61148c2bf7ce2fa7216907f9fadd4ff42d5256d0
- Math Research 1: /workspace/pqc-gemini-request-intake-20261105-1513/math1.md
ccbb3f47bd0116d975907d48dfc7bec8b223853ce663d0a770204668722c8b64
- Math Research 2: /workspace/pqc-gemini-request-intake-20261105-1513/math2.md
4a89f647f05bf906442e4c239d9646a1d9b57b39a49c6e7d4a1f99de04a135bf
- Physics Research 1: /workspace/pqc-gemini-request-intake-20261105-1513/physics1.md
a4d6d42ff3fba925e36876cf8c25336382b8a6fca371a89b55718df8584743ce
- Physics Research 2: /workspace/pqc-gemini-request-intake-20261105-1513/physics2.md
908aaa0d4405c3f65e4ec0adebf5b6f18536bd165454156215f6f7154317c2e6
- Eigenvalues and matrix: /workspace/pqc-gemini-request-intake-20261105-1513/eigen.md
bde19232007b366af78035861c93123ec23215ad3994f9ce97b246d5b59e5790
- Dataset Filtration: /workspace/pqc-gemini-request-intake-20261105-1513/dataset.md
7a331bfebc3448f4be0c5190df544a7d6e293c0c9d51b71ce47714cfa5a27d80
- Agent: /workspace/pqc-gemini-request-intake-20261105-1513/agent.md
7bcfb60f24fb56c85d060181ebf1e8db56c09331a38f16d72fa9ac4933d2c7be
- 539 Labs Ops additive correction:
/workspace/pqc-gemini-request-intake-20261105-1513/labs539_ADDITIVE_CORRECTION_T3REF.md
c8e064c768e76e1766af66ea07f4da1fedaf6d2e1d95b6491a9db0c3c3e2fe45
- Rule, Primary autonomy: /workspace/quantum-comp/reports/OWNER_WRITE_CALL_53991ebf_PRIMARY_OPERATING_RULE_ALL_TWELVE_2026-11-05.md
bcc3ced0f1dfe15560e0bdb2e45bb35f4b1db26dcfb439e647a98c12fc5bc9a9

- Rule, Full SM objective: /workspace/quantum-comp/reports/OWNER_DIRECTIVE_GR_EXTENSION_FULL_STANDARD_MODEL_RECOVERY_2026-11-05.md 6237cc716829c2838368099382c7ecdebfed46895491f8ecb37ee5cc167c815e
- Rule, 80% stop: /workspace/quantum-comp/reports/OWNER_DIRECTIVE_WEEKLY_USAGE_80_STOP_AND_FINAL_REPORT_2026-11-05.md 51ff16771c69d877675ac7c59e269d76444180784e81842c7818029387d1747e
- Rule, Gemini ask-then-verify: /workspace/quantum-comp/reports/OWNER_DIRECTIVE_GEMINI_COMPUTATION_TOKEN_PRESERVATION_VERIFY_2026-11-05.md 3921181508acede9740ba9892b35cb1e379d25ccc0af4f601cbe803a01b68b67
- Rule, Gemini file loop: /workspace/quantum-comp/reports/OWNER_DIRECTIVE_GEMINI_FILE_LOOP_VERIFICATION_ONLY_2026-11-05.md a29f87d93fd75a9a34dc7acce66ab5474968180bb532d3a8d11804d6539253be
- Rule, Standing rules index (file loop): /workspace/quantum-comp/reports/STANDING_RULES_INDEX_GEMINI_FILE_LOOP_UPDATE_2026-11-05.md 3abbfb9e91650e5f3dd0fccf50c0d7dc67499dfb48ad82805e1611151b5e073c
- Companion (prior 77-page Q&A compendium PDF, separate file, unchanged): /workspace/quantum-comp/reports/GEMINI_COMPENDIUM_AGENT_QUESTIONS_PROOF_REQUESTS_CONSOLIDATED_2026-11-05.pdf 3eacced22f4f901cfcd110418f859cc4996da1f370fe8d697f8f9d11edf1e6a8

END OF DRAFT PACKET. Draft request, not validated findings. No external Gemini submission or transfer was performed.